Customer Shopping Behavior Analysis

0. Customer_Shopping_Behavior.csv:

1. Linhas: *3900*
2. Colunas: *18*
3. Key Features:
- ***Customer Identification:***
  - *Customer ID (Unique identifier for each customer).*
- ***Customer demographics:***
  - *Age (Customer's age);*
  - *Gender (Customer's gender);*
  - *Location (State or region where the customer resides (in the US, in the case of this dataset)).*
- ***Purchase and product details:***
  - *Item Purchased (Specific item purchased);*
  - Category (General category of the item);
  - *Purchase Amount (USD) (Transaction value in dollars);*
  - *Size (Size of the item purchased);*
  - *Color (Item color);*
  - *Season (The season in which the purchase was made).*
- ***Experience:***
  - *Review Rating (The rating given by the customer to the product).*
- ***Loyalty:***
  - *Subscription Status (Whether the customer has an active subscription with the store);*
  - *Previous Purchases (Number of purchases this customer has made previously);*
  - *Frequency of Purchases (How often the customer usually buys).*
- ***Payment:***
  - *Shipping Type (Shipping method chosen);*
  - *Discount Applied (Indicates if any discount was applied to the purchase);*
  - *Promo Code Used (Indicates whether a promotional code was used);*
  - *Payment Method (Payment method used).*

4. Null values: *37 in Review Rating.*

1. ETL - Extract, Transform, and Load (Python):

1. Imports: The pandas, sqlalchemy, pymysql, and ipykernel libraries were installed, and the first and second mentioned were imported;

2. Loading the dataset: After loading the dataset from the data/raw folder, df.info() and df.head() were used to display information and show the first five rows of the dataset, respectively;

3. Data preprocessing: Basic statistics were explored using df.describe(include='all') to also include categorical columns; it was verified that the Rating Review column had 37 rows with null values, which were replaced by the median of their respective category; the column names were standardized, changing uppercase letters to lowercase and spaces to "_" and removing (USD) from Purchase Amount;

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

4. Manipulating columns: The age_range and purchase_frequency_days columns were created to standardize ages into four groups and show the frequency of purchases in days; it was verified that the discount_applied and promotional_code_used columns were the same, therefore the second one was removed;

5. Exporting the CSV: The cleaned and transformed dataset was exported to the data/processed column;

6. Connecting to MySQL: A connection was established between Python and MySQL, with the table being loaded for analysis in SQL, which was performed in phpMyAdmin.

2. Data Analysis (SQL):

1. *What is the total revenue generated by male customers compared to female customers?*

gender	purchase_sum 1
Male	157890
Female	75191

2. *Which customers received a discount but still spent more than the average purchase amount?*

customer_id	purchase_amount
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69
55	94
57	73
58	64
60	79
62	68

Total de clientes: 839

3. *What are the 5 products with the highest average rating?*

item_purchased	review_mean
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. *Comparing average purchase values between standard and express shipping.*

shipping_type	purchase_mean 1
Express	60.48
Standard	58.46

5. *Did subscribers spend more? Comparing average spending and total revenue between subscribers and non-subscribers.*

subscription_status	customer_total	purchase_mean 2	purchase_total 1
No	2847	59.87	170436.00
Yes	1053	59.49	62645.00

Non-subscribing customers have a higher average spend and total revenue than subscribing customers.

6. *What are the 5 products with the highest percentage of purchases with discounts applied?*

item_purchased	discount_applied 1
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. *Segmenting customers into New, Returning, and Loyal based on the total number of previous purchases and show the count for each segment.*

client_type	client_count 1
Loyal	3116
Returning	701
New	83

8. *What are the 3 most purchased products in each category?*

category	item_purchased	purchase_total	ranking
Accessories	Belt	161	3
Accessories	Jewelry	171	1
Accessories	Sunglasses	161	2
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Coat	161	2
Outerwear	Jacket	163	1

9. *Are customers who buy repeatedly (more than 5 previous purchases) also more likely to repeat?*

customers 1	subscription_status
2518	No
958	Yes

Non-Subscribers: 2,518
Subscribers: 958
Total: 3,476
Subscription Rate: Only 27.5% of this group are subscribers

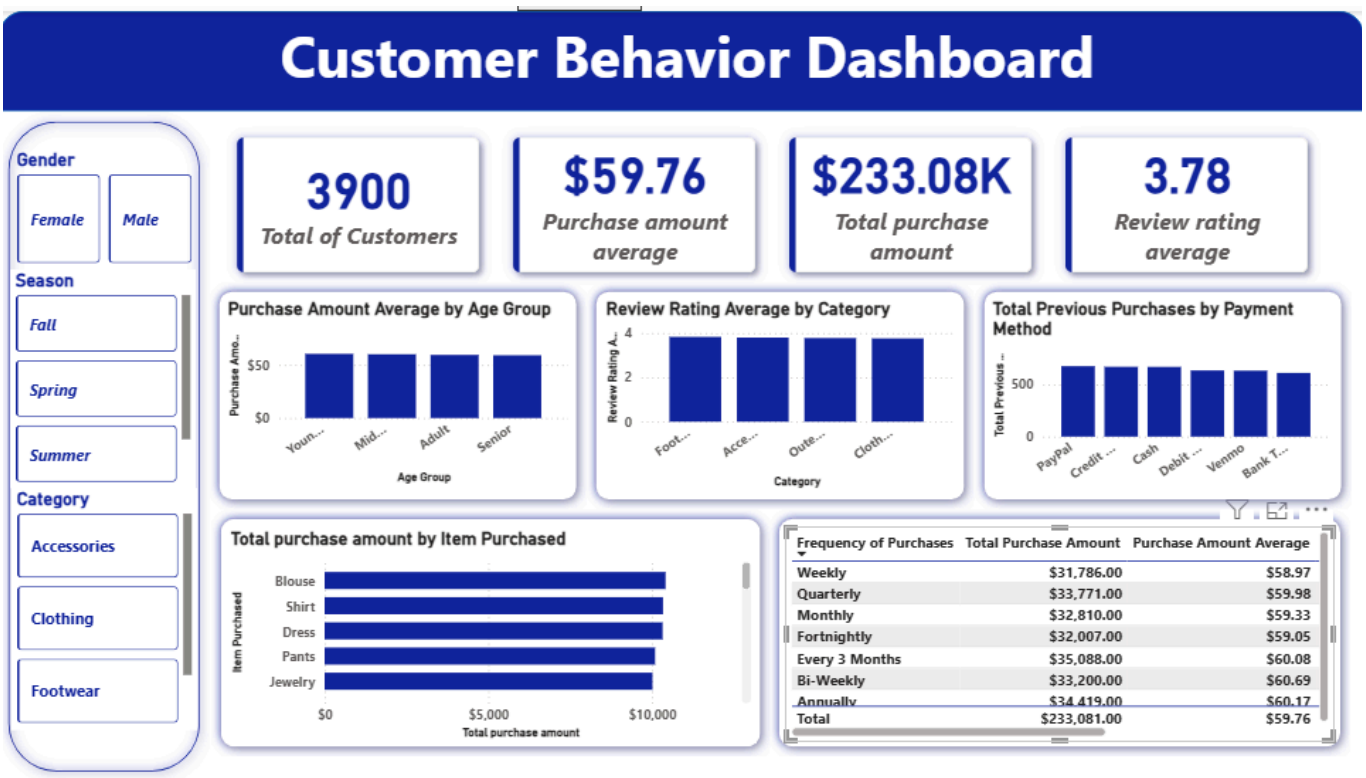
Therefore, having more than 5 previous purchases does not increase the likelihood of signing up.

10. *What is the contribution of each age group to revenue?*

age_group	purchase_amount 1
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

3. Dashboard (Power BI):

Interactive dashboard built in Power BI:



4. Business Recommendations:

1. Male customers account for 67.74% of revenue. It is recommended to expand the female customer portfolio and invest in targeted campaigns for this audience, seeking a more balanced distribution;

2. Only 21.51% of customers received discounts above the media rate, indicating that discounts are not increasing the average order value. It is recommended to adopt progressive discounts or discounts conditional on a minimum purchase amount;

3. No product achieved an average score higher than 4.0. It is recommended to prioritize reviewing suppliers or replacing products;

4. 79.90% of customers are loyal, but only 25.7% subscribe—and these represent only 26.87% of revenue. There is a clear gap between loyalty and conversion. It is recommended to make subscription benefits more attractive and communicate the assets of the loyal customer base.